

Unit 8 | Chapter 15: The "Future" (Emerging Tech: AI, AR/VR, IoT)

Learning Targets:	
❖ Technical: I can define and differentiate between Artificial Intelligence (AI), Augmented Reality (AR), and the Internet of Things (IoT). (NV 7.2.1) ❖ Technical: I can analyze the impact of machine learning and large language models (LLMs) on modern web development workflows. (NV 7.2.1) ❖ Technical: I can describe the hardware and software requirements for immersive web experiences (WebXR). (NV 7.2.1) ❖ Technical: I can explain the security and privacy risks associated with a hyper-connected IoT ecosystem. (NV 3.2.8) ❖ Employability: I can demonstrate analytical reasoning and cognitive flexibility to predict how emerging technologies will disrupt current labor market trends. (WRS 1.2.3) ❖ Employability: I can demonstrate probity and ethical adherence by evaluating the algorithmic bias and intellectual property concerns in AI-generated content. (WRS 1.1.2)	
Compelling Question:	
How will the convergence of AI, immersive reality, and ubiquitous connectivity redefine what it means to be a "Web Developer" in the coming decade?	
Score	Performance Indicators
4.0	I can: ☐ Develop a prototype or detailed technical proposal for a web application that integrates an AI API or WebXR components, demonstrating advanced analytical reasoning and systematic resolution. (NV 7.2.1, WRS 1.2.3) ☐ Critically analyze the ethical implications of "Black Box" algorithms on marginalized communities, demonstrating profound cultural competency and empathy. (WRS 1.1.2, 1.1.5) ☐ Display exceptional cognitive flexibility by successfully adapting a legacy project to incorporate one emerging technology feature. (WRS 1.2.3)
3.5	In addition to score 3.0 performance, partial success at score 4.0 content.
3.0	I can: ☐ Explain the Request/Response cycle of an AI API (like Gemini or OpenAI) and how it differs from a traditional REST API. (NV 7.2.1) ☐ Identify the ethical risks of AI (Hallucinations, Bias, Plagiarism) and apply probity when using AI tools for coding assistance. (WRS 1.1.2) ☐ Describe the practical applications of IoT in smart cities and homes, focusing 

	on the professional decorum required to protect user privacy. (WRS 1.1.4) ☐ Demonstrate intrinsic motivation and punctuality by completing research summaries on emerging tech on schedule. (WRS 1.1.1)
2.5	No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.
2.0	I can ☐ Define common terminology and identify syntax: ☐ Artificial Intelligence, Machine Learning, LLM, Prompt Engineering, Augmented Reality (AR), Virtual Reality (VR), WebXR, IoT, Algorithmic Bias, Edge Computing. (NV 7.2.1) ☐ Identify the importance of punctuality (WRS 1.1.1), probity in AI ethics (WRS 1.1.2), professional decorum (WRS 1.1.4), inclusive collaboration (WRS 1.1.5), and applying analytical reasoning to future tech trends (WRS 1.2.3).
1.5	Partial success at score 2.0 content, and major errors or omissions at 3.0.
1.0	With help, partial success at score 2.0 and score 3.0 content.
0.5	With help, partial success at score 2.0 but not at score 3.0 content.
0.0	Even with help, no success or attempt made.